Drug Resistance Mutations Affect on HLA Binding in HIV-1 Epitopes

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Chapter 1

Introduction & Background

Human Immunodeficiency Virus Type 1 (HIV-1), a virus transmitted by body fluids, capable of shutting down the host immune system if not treated immediately [23]. As of 2020, an estimated 37 million people around the globe suffer from HIV-1 [28]. Furthermore, HIV-1 continues to spread, where it is estimated an additional 1.7 million global citizens acquire HIV-1 on average each year [28]. Despite having been discovered over 4 decades ago in 1981 [25], HIV-1, the cause of Acquired Immune Deficiency Syndrome (AIDS), remains an incurable disease and evades any form of vaccination or cure [28] where the best treatment option for patients of HIV-1 is to undergo antiretroviral therapy (ART) treatment [17]. With ART, HIV-1 associated mortality rates have been successfully reduced ultimately saving millions, and changing AIDS from a deadly disease, to one where the patients may typically live a full life [28]. However, it is critical to accept ART not as a complete solution, rather, as a temporary solution. ART itself remains an imperfect treatment with problems ranging from rebound due to patients not fulfilling the required periodic treatment plan [2], to constrained availability of treatment [22], to problematic cost of treatment [10, 25].

1.1 Drug Resistance

Amongst these problems, however, there underlies a biologically critical change HIV-1 is undergoing common to many pathogens where a treatment is not potent enough to cure is the gold standard and thus deterring the progress of ART. Specifically, in recent years, with advent of ART, HIV-1, has been slowly but surely developing resistance to many modern ART treatment related drugs [3, 4, 7].

To elaborate on HIV-1 drug resistant mutations (DRMs), it is typical that a single amino acid variation coded for in the HIV-1 genome is not enough to induce resistance towards antiretroviral drugs (ARDs) [3]. Furthermore, within a given drug classification, it is rather common to see cross-resistance where the increased resistance towards one particular drug increases the resistance

towards another [3]. Despite such cross-resistance prevalence within the same drug class, the same cannot be said across different drug classes where current research seems to indicate that it is more difficult [11] for such cross-resistances, however, not impossible [3, 13, 24]. This lower prevalence of cross-class drug resistance is a key reason for the use of multiple ARDs of differing classes during modern ART treatment plans [6]. Notably, such DRMs seem to typically result in a fitness penalty to the host strain of the DRMs [3].

1.2 Immune Escape

HIV-1 is fascinating in that it is one of the few viruses in modern day that both is highly prevalent in the population, but also eludes cures. Specifically, Cytotoxic T Cell (CTL), also known as CD8+ T cells, are well known to be a host's main antagonist to theHIV-1 virus [5]. Then, it goes to reason, that HIV-1, with a relatively unstable genome capable of high rates of proliferation [21], since it's initial entry into the human population, has developed various mechanisms of circumvent, or escape, an effective CTL response. One such mechanism is via certain specific immune escape mutations (IEMs) that result in the suppressing of displaying specifically HIV-1 associated antigens thus avoiding apoptosis via natural killer T cells [27]. Similar DRMs, IEMs such IEMs also seem to introduce a fitness penalty to the HIV-1 strain [12, 31].

1.3 Motivation for Study

Both mutations, independent of each other are relatively thoroughly studied. However, at the time of writing, there exists a minimal amount of studies looking into the intersection between both DRMs and IEMs. In other words, since both forms of mutations incur fitness penalties, an area of interest may be how DRMs affect the mutation based immune escape process via epitopic modifications. The few studies emphasizing such a topic discusses various DRMs that both corroborate the increase of immunogenicity of HIV-1 [1, 15], and a decrease in immunogenicity, depending on varying factors [16, 18]. The results of these studies are reasonable as DRMs occurring within epitope regions, i.e., regions that affect the presentation via human leukocyte antigen class I (HLA-I) may modulate the fitness penalty due to directly affecting a immune escape mechanism. This study hopes to extend the knowledge base surrounding such interplay between DRMs and IEMs as such knowledge may serve foundational to improving treatment regiment by selecting drugs that produce strong fitness penalties and the future development of HIV-1 vaccines, as explained by previous studies [16, 18].

Chapter 2

Materials & Methods

2.1 Materials

Analysis of the interplay between DRMs, and IEMs were performed upon datasets upon the Los Alamos HIV databases (LANL-HIV) [14] and the Stanford HIV Drug Resistance Mutation Database (DRM-DB) [20]. NetMHCPan version 4.1 and an in-house python script were used in combination for accommodating large data throughput [19]. Within the Python script, SciPy and Pandas data processing libraries were used for the analysis [26, 29]. Figures were generated with use of Matplotlib and Seaborn [8, 30].

The LANL-HIV dataset of known HIV-1 CTL epitopes including the peptide sequence and position relative to the gene. Datasets were individually downloaded by querying for integrase, p24, protease, and Reverse Transcriptase (RT) while all other search parameters were left as the most broad casting option i.e., selecting the "ALL" option when present. The DRM-DB amongst other tools, and other datasets, contains a scoring mechanism for known drug resistance inducing mutations in the HIV-1 genome [20]. Spreadsheets of such scoring for the different drugs with respect to a given single mutation were downloaded. NetMHCPan version 4.1 was used to predict binding against different HLA-I alleles [19]. Data ingress and egress, manipulation and vectorized operations were performed by Pandas and the SciPy library was used for statistical associations [26, 29]. Lastly, Seaborn was used for the automated generation and regression figures by making use of Matplotlib [8, 30].

2.2 Methods

All aforementioned datasets were imported from respective comma separated values (CSV) sheets downloaded from their respective databases. This was then followed by the generation of new epitopes based off the set of known epitopes, and the set of drug resistant mutations. All epitopes, natural or generated were then run through NetMHCPan 4.1 to predict their binding level

towards various HLA-I alleles. Such binding levels were fit into categories. Hypothetical epitopes were generated from the naturally occurring epitopes (N-EPIs). All generated epitopes were cleaned and filtered to increase the number of interpretable natural epitopes and reduce reduce computation time respectively before running the dataset in NetMHCPan [19]. DRMs and their associated resistance scores were summed in accordance to the guidelines of the DRM-DB. The resistance scores and DRM scores were then categorized according to their respective guidelines from DRM-DB. Association between the presence of DRMs and the binding level with various HLA-I alleles were tested and p-values were aggregated.

The first step involved downloading the data and parsing the data into Python. Data was procured from both LANL-HIV and DRM-DB as described in the materials section. DRM were further parsed and merged into a large, master spreadsheet where the drugs irrelevant to a mutation were set to have a DRM of 0. Similarly, known N-EPIs were also parsed. Unused columns of metadata were dropped and epitopes with the exact identical sequences were dropped immediately as to ensure that any associations found would be purely based on their occurrence in a epitope, and not biased by the frequency in a population. Furthermore, the LANL-HIV contained synthetic epitopes created by concatenating two epitopes together. Such epitopes did not have well-defined bounds and thus, also did not have well-defined DRMs [14]. Such junction-based epitopes had their junctions removed and the original epitope restored by selecting the portion of the junction epitope based on the annotated bound.

The unique set of naturally occurring HIV epitopes then had variants generated naively simply by generating all possible different combinations of mutations that could be added to a naturally occurring epitope was created. These epitopes were then added to a new flat database with their original DRMs, i.e., mutations that existed before any modification, and the added mutations. Notably, due to some epitopes having many possible combination of DRMs, constraints were added such that generated epitopes that resulted in a DRM score lower than 15 was removed. This corresponds to DRM-DBs categorization of having none, or potentially a very low level drug resistance (Table 2.2). Additionally, some N-EPI had many peptide positions with known DRMs and thus, a ranked threshold was applied where the DRM scores were used to rank the generated epitopes and such epitopes that ranked higher than 1000 were removed. I.e., the top 1000 highest DRM scoring generated epitopes were kept.

Then, the produced aggregate of both N-EPIs and the generated epitopes had binding levels with all HLA-I alleles that were recognized by NetMHCPan 4.1 (table 2.1) predicted and cataloged [19]. The resulting database had DRM scores and eluted ligand percentile rank (%EL-Rank) [9] calculated and mapped according to table 2.2 and to table 2.3 respectively (for more information on this metric, see original software paper for version 4.0) [9, 19]. Furthermore, changes in binding level were calculated by comparing the N-EPIs with the generated ones.

Lastly, all combinations between all sets of DRMs, and the empty set of DRMs, i.e., the lack of DRMs, were tested for association with all different levels

Supported HLA Variants
HLA-A02:01
HLA-A01:01
HLA-A03:01
HLA-A24:02
HLA-A26:01
HLA-B07:02
HLA-B08:01
HLA-B27:05
HLA-B39:01
HLA-B40:01
HLA-B58:01
HLA-B15:01

Table 2.1: The supported list of HLA alleles for which NetMHCPan is capable of predicting %EL-Rank.

Resistance Category	DRM Score (Inclusive)
No drug resistance	0 to 9
Potential low-level drug resistance	10 to 14
Low-level drug resistance	15 to 29
Intermediate resistance	30 to 59
High-level resistance	60 and above

Table 2.2: The categorical levels of drug resistance based on the DRM score assigned for an entire sequence as defined by DRM-DB.

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High-level resistance	60 and above

Table 2.3: The categorical levels of HLA binding based on the percentile rank predicted by NetMHCPan 4.1 where both the metric and the thresholds were advised by the original paper and authors of the software [19].

of binding, that is to say, these combinations of DRMs were tested against the occurrences of weak, strong, and no binding level epitopes. The Fisher's exact test was used due to the relatively small sample count of epitopes with the same set of DRMs with unique epitopes. This process generated a list of different p-values for each possible combination. This list of p-values was then filtered by a $\alpha = 0.05$.

Resulting tables were generated via Pandas and resulting figures were generated with Seaborn and Matplotlib [8, 26, 30].

Chapter 3

Results

3.1 Binding Rank in Association with Drug Resistance

Chapter 4

Discussion

Chapter 5

Acronyms

%EL-Rank eluted ligand percentile rank

AIDS Acquired Immune Deficiency Syndrome

ARD antiretroviral drug

ART antiretroviral therapy

CSV comma separated values

CTL Cytotoxic T Cell

DRM drug resistant mutation

DRM-DB Stanford HIV Drug Resistance Mutation Database

HIV-1 Human Immunodeficiency Virus Type 1

HLA-I human leukocyte antigen class I

IEM immune escape mutation

LANL-HIV Los Alamos HIV databases

N-EPI naturally occurring epitope

RT Reverse Transcriptase

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